

Hierarchical & Density-Based Clustering

Numerical Worksheet

INT396 • Unsupervised Learning • Unit III

School of Computer Science & Engineering

Solutions

Distances are Euclidean, $d(p, q) = \sqrt{\sum_i (p_i - q_i)^2}$. Round to four decimal places unless stated otherwise.

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Formulas

Agglomerative algorithm start with n singleton clusters and the $n \times n$ distance matrix. Repeat $n - 1$ times: find the smallest entry, merge that pair at height equal to that entry, replace the two rows/columns by one using the linkage rule. After m merges there are $n - m$ clusters.

Candidate counts

$$\text{first agglomerative step: } \binom{n}{2} = \frac{n(n-1)}{2}, \quad \text{first divisive step: } 2^{n-1} - 1.$$

Linkage rules

$$d_{\text{single}}(A, B) = \min_{a \in A, b \in B} d(a, b), \quad d_{\text{complete}}(A, B) = \max_{a \in A, b \in B} d(a, b),$$

$$d_{\text{avg}}(A, B) = \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} d(a, b)$$

$$\Delta_{\text{ESS}}(A, B) = \frac{|A||B|}{|A| + |B|} \|\mu_A - \mu_B\|^2, \quad d_{\text{ward}}(A, B) = \sqrt{2 \Delta_{\text{ESS}}(A, B)}$$

where

$|A|$ number of points in cluster A

μ_A centroid of cluster A

Updating after the merge $A \cup B$ distance to another cluster C :

single: $\min(d(A, C), d(B, C))$, complete: $\max(d(A, C), d(B, C))$,

average: $\frac{|A| d(A, C) + |B| d(B, C)}{|A| + |B|}$, Ward: $\mu_{A \cup B} = \frac{|A| \mu_A + |B| \mu_B}{|A| + |B|}$, $|A \cup B| = |A| + |B|$.

Cutting a dendrogram

$$\# \text{clusters at height } h = n - \# \{ \text{merges with height } \leq h \}$$

With sorted merge heights $h_1 \leq \dots \leq h_{n-1}$, exactly k clusters for $h \in [h_{n-k}, h_{n-k+1})$.

Cophenetic distance $d_{\text{coph}}(x_i, x_j)$ = height of the merge at which x_i and x_j first share a cluster. CPCC = Pearson correlation between the $\binom{n}{2}$ true distances and the $\binom{n}{2}$ cophenetic distances:

$$\text{CPCC} = \frac{\sum (t_i - \bar{t})(c_i - \bar{c})}{\sqrt{\sum (t_i - \bar{t})^2 \sum (c_i - \bar{c})^2}}$$

DBSCAN $N_\varepsilon(p) = \{q : d(p, q) \leq \varepsilon\}$, with p included.

- core: $|N_\varepsilon(p)| \geq \text{MINPTS}$
- border: not core, and within ε of a core point
- noise: neither

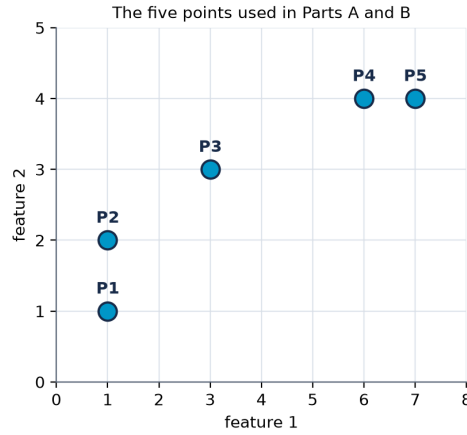
Core points within ε of each other are in the same cluster; border points join a core point that reaches them. Smallest ε making p core = distance from p to its $(\text{MINPTS} - 1)$ -th nearest other point (k -distance).

Costs distance matrix: $\binom{n}{2}$ distances, $8\binom{n}{2}$ bytes in double precision. k-Means: $n \times k \times$ (iterations) point-to-centroid distances.

1 Agglomerative vs. divisive clustering

Worked example 1.1 — *full single-linkage run*

$P_1(1, 1)$, $P_2(1, 2)$, $P_3(3, 3)$, $P_4(6, 4)$, $P_5(7, 4)$. $n = 5$, so $\binom{5}{2} = 10$ distances and $n - 1 = 4$ merges.



All ten distances

$$d(P_1, P_2) = \sqrt{(1-1)^2 + (2-1)^2} = \sqrt{0+1} = \sqrt{1} = 1.0000$$

$$d(P_1, P_3) = \sqrt{(3-1)^2 + (3-1)^2} = \sqrt{4+4} = \sqrt{8} = 2.8284$$

$$d(P_1, P_4) = \sqrt{(6-1)^2 + (4-1)^2} = \sqrt{25+9} = \sqrt{34} = 5.8310$$

$$d(P_1, P_5) = \sqrt{(7-1)^2 + (4-1)^2} = \sqrt{36+9} = \sqrt{45} = 6.7082$$

$$d(P_2, P_3) = \sqrt{(3-1)^2 + (3-2)^2} = \sqrt{4+1} = \sqrt{5} = 2.2361$$

$$d(P_2, P_4) = \sqrt{(6-1)^2 + (4-2)^2} = \sqrt{25+4} = \sqrt{29} = 5.3852$$

$$d(P_2, P_5) = \sqrt{(7-1)^2 + (4-2)^2} = \sqrt{36+4} = \sqrt{40} = 6.3246$$

$$d(P_3, P_4) = \sqrt{(6-3)^2 + (4-3)^2} = \sqrt{9+1} = \sqrt{10} = 3.1623$$

$$d(P_3, P_5) = \sqrt{(7-3)^2 + (4-3)^2} = \sqrt{16+1} = \sqrt{17} = 4.1231$$

$$d(P_4, P_5) = \sqrt{(7-6)^2 + (4-4)^2} = \sqrt{1+0} = \sqrt{1} = 1.0000$$

Distance matrix

	P_1	P_2	P_3	P_4	P_5
P_1	—	1.0000	2.8284	5.8310	6.7082
P_2		—	2.2361	5.3852	6.3246
P_3			—	3.1623	4.1231
P_4				—	1.0000
P_5					—

Merge 1

Smallest off-diagonal entry: 1.0000, attained by (P_1, P_2) and (P_4, P_5) . Take (P_1, P_2) first. Merge $\{P_1, P_2\}$ at height 1.0000. Clusters now: $\{P_1, P_2\}$, $\{P_3\}$, $\{P_4\}$, $\{P_5\}$ ($5 - 1 = 4$). New row for $\{P_1, P_2\}$ (single linkage: minimum of the two old rows):

$$d(\{P_1, P_2\}, P_3) = \min(d(P_1, P_3), d(P_2, P_3)) = \min(2.8284, 2.2361) = 2.2361$$

$$d(\{P_1, P_2\}, P_4) = \min(d(P_1, P_4), d(P_2, P_4)) = \min(5.8310, 5.3852) = 5.3852$$

$$d(\{P_1, P_2\}, P_5) = \min(d(P_1, P_5), d(P_2, P_5)) = \min(6.7082, 6.3246) = 6.3246$$

Unchanged: $d(P_3, P_4) = 3.1623$, $d(P_3, P_5) = 4.1231$, $d(P_4, P_5) = 1.0000$.

	$\{P_1, P_2\}$	P_3	P_4	P_5
$\{P_1, P_2\}$	—	2.2361	5.3852	6.3246
P_3		—	3.1623	4.1231
P_4			—	1.0000
P_5				—

Merge 2

Entries: 2.2361, 5.3852, 6.3246, 3.1623, 4.1231, 1.0000. Smallest: 1.0000 at (P_4, P_5) . Merge $\{P_4, P_5\}$ at height 1.0000. Clusters: $\{P_1, P_2\}$, $\{P_3\}$, $\{P_4, P_5\}$ ($5 - 2 = 3$).

$$d(\{P_4, P_5\}, \{P_1, P_2\}) = \min(d(P_4, \{P_1, P_2\}), d(P_5, \{P_1, P_2\}))$$

$$= \min(5.3852, 6.3246) = 5.3852$$

$$d(\{P_4, P_5\}, P_3) = \min(d(P_4, P_3), d(P_5, P_3)) = \min(3.1623, 4.1231) = 3.1623$$

Unchanged: $d(\{P_1, P_2\}, P_3) = 2.2361$.

	$\{P_1, P_2\}$	P_3	$\{P_4, P_5\}$
$\{P_1, P_2\}$	—	2.2361	5.3852
P_3		—	3.1623
$\{P_4, P_5\}$			—

Merge 3

Entries: 2.2361, 5.3852, 3.1623. Smallest: 2.2361 at $(\{P_1, P_2\}, P_3)$. Merge $\{P_1, P_2, P_3\}$ at height 2.2361. Clusters: $\{P_1, P_2, P_3\}$, $\{P_4, P_5\}$ ($5 - 3 = 2$).

$$d(\{P_1, P_2, P_3\}, \{P_4, P_5\}) = \min(d(\{P_1, P_2\}, \{P_4, P_5\}), d(P_3, \{P_4, P_5\}))$$

$$= \min(5.3852, 3.1623) = 3.1623$$

	$\{P_1, P_2, P_3\}$	$\{P_4, P_5\}$
$\{P_1, P_2, P_3\}$	—	3.1623
$\{P_4, P_5\}$		—

Merge 4

Only one entry, 3.1623. Merge $\{P_1, P_2, P_3, P_4, P_5\}$ at height 3.1623. Clusters: 1 ($5 - 4 = 1$). Stop.

Merge table

$(P_1, P_2)@1.0000$, $(P_4, P_5)@1.0000$, $(P_3, \{P_1, P_2\})@2.2361$, $(\{P_1, P_2, P_3\}, \{P_4, P_5\})@3.1623$

Worked example 1.2 — *candidate counts* $n = 6$

$$\binom{6}{2} = \frac{6 \times 5}{2} = \frac{30}{2} = 15 \quad 2^{6-1} - 1 = 2^5 - 1 = 32 - 1 = 31$$

 $n = 20$

$$\binom{20}{2} = \frac{20 \times 19}{2} = \frac{380}{2} = 190 \quad 2^{20-1} - 1 = 2^{19} - 1 = 524,288 - 1 = 524,287$$

Questions

Q1.1 Points 1, 2, 4, 8, 13. Write the distance matrix and run single-linkage agglomeration to completion, showing the smallest entry, the merge, its height and the reduced matrix at every step.

**Solution.**

	1	2	4	8	13
1	—	1	3	7	12
2		—	2	6	11
4			—	4	9
8				—	5
13					—

Merge 1: {1, 2} at 1

$d(\{1, 2\}, 4) = \min(3, 2) = 2$, $d(\{1, 2\}, 8) = \min(7, 6) = 6$, $d(\{1, 2\}, 13) = \min(12, 11) = 11$;
 $d(4, 8) = 4$, $d(4, 13) = 9$, $d(8, 13) = 5$.

Merge 2: {1, 2, 4} at 2

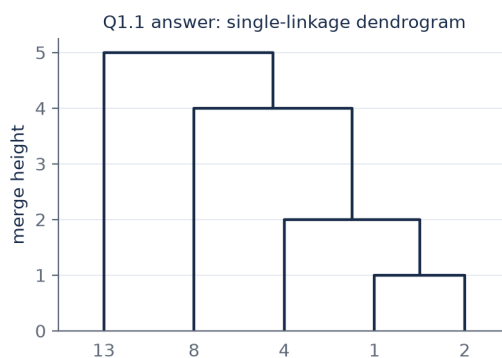
$d(\{1, 2, 4\}, 8) = \min(6, 4) = 4$, $d(\{1, 2, 4\}, 13) = \min(11, 9) = 9$; $d(8, 13) = 5$.

Merge 3: {1, 2, 4, 8} at 4

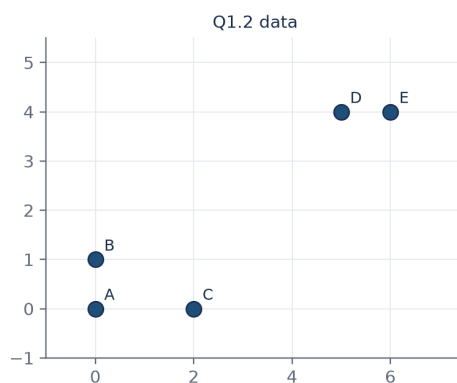
$d(\{1, 2, 4, 8\}, 13) = \min(9, 5) = 5$.

Merge 4: all at 5

$$(1, 2)@1, \quad (4, \{1, 2\})@2, \quad (8, \{1, 2, 4\})@4, \quad (13, \{1, 2, 4, 8\})@5$$



Q1.2 $A(0, 0)$, $B(0, 1)$, $C(2, 0)$, $D(5, 4)$, $E(6, 4)$. Compute the distance matrix and run complete-linkage agglomeration to completion, showing the reduced matrix after each merge.



Solution.

	A	B	C	D	E
A	—	1.0000	2.0000	6.4031	7.2111
B		—	2.2361	5.8310	6.7082
C			—	5.0000	5.6569
D				—	1.0000
E					—

Merges 1, 2: $\{A, B\}$ and $\{D, E\}$ at **1.0000**

$$d(\{A, B\}, C) = \max(2.0000, 2.2361) = 2.2361$$

$$d(C, \{D, E\}) = \max(5.0000, 5.6569) = 5.6569$$

$$d(\{A, B\}, \{D, E\}) = \max(6.4031, 7.2111, 5.8310, 6.7082) = 7.2111$$

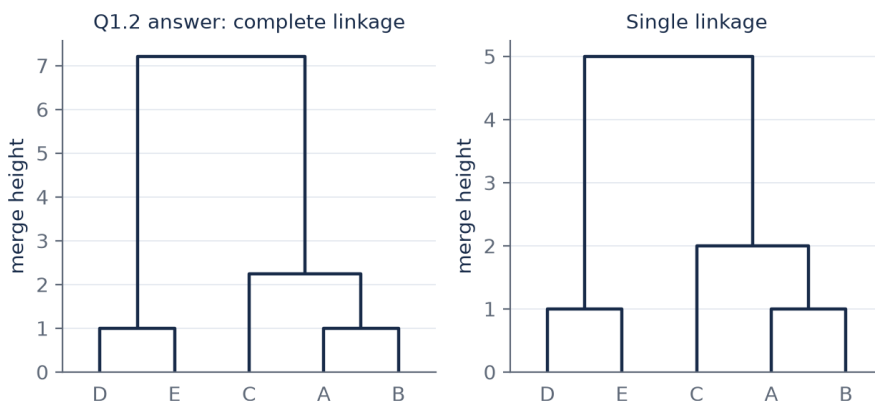
	$\{A, B\}$	C	$\{D, E\}$
$\{A, B\}$	—	2.2361	7.2111
C		—	5.6569
$\{D, E\}$			—

Merge 3: C into $\{A, B\}$ at **2.2361**

$$d(\{A, B, C\}, \{D, E\}) = \max(7.2111, 5.6569) = 7.2111.$$

Merge 4: all at **7.2111**

$$(A, B)@1.0000, \quad (D, E)@1.0000, \quad (C, \{A, B\})@2.2361, \quad (\{A, B, C\}, \{D, E\})@7.2111$$



Q1.3 (a) For $n = 9$: number of merges in a full run; clusters after 3 merges; merges needed to reach $k = 3$. **(b)** For $n = 7$ and $n = 12$: $\binom{n}{2}$, $2^{n-1} - 1$, and their ratio.

Solution.

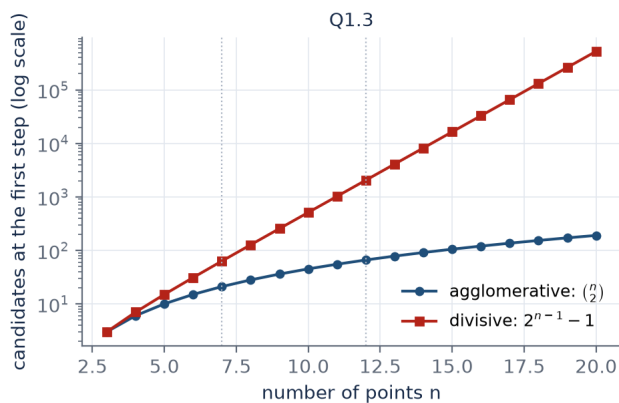
(a)

Merges = $n - 1 = 8$. After 3 merges: $9 - 3 = 6$ clusters. For $k = 3$: $9 - 3 = 6$ merges.

(b)

$$n = 7: \binom{7}{2} = 21, \quad 2^6 - 1 = 63, \quad 63/21 = \mathbf{3.00}$$

$$n = 12: \binom{12}{2} = 66, \quad 2^{11} - 1 = 2047, \quad 2047/66 = \mathbf{31.02}$$



2 Linkage methods

Worked example 2.1 — one merge, four linkages

Cluster $A = \{P_1, P_2\}$, $P_1(1, 1)$, $P_2(1, 2)$; cluster $B = \{P_3\}$, $P_3(3, 3)$. $|A| = 2$, $|B| = 1$, so $|A| \times |B| = 2$ cross-pairs.

Cross-pair distances

$$d(P_1, P_3) = \sqrt{(3-1)^2 + (3-1)^2} = \sqrt{4+4} = \sqrt{8} = 2.8284$$

$$d(P_2, P_3) = \sqrt{(3-1)^2 + (3-2)^2} = \sqrt{4+1} = \sqrt{5} = 2.2361$$

Single

$$d_{\text{single}}(A, B) = \min(2.8284, 2.2361) = \mathbf{2.2361}$$

Complete

$$d_{\text{complete}}(A, B) = \max(2.8284, 2.2361) = \mathbf{2.8284}$$

Average

$$d_{\text{avg}}(A, B) = \frac{1}{2 \times 1} (2.8284 + 2.2361) = \frac{5.0645}{2} = \mathbf{2.5322}$$

Ward — centroids

$$\mu_A = \left(\frac{1+1}{2}, \frac{1+2}{2} \right) = \left(\frac{2}{2}, \frac{3}{2} \right) = (1, 1.5), \quad \mu_B = (3, 3)$$

Ward — squared centroid distance

$$\|\mu_A - \mu_B\|^2 = (1-3)^2 + (1.5-3)^2 = (-2)^2 + (-1.5)^2 = 4 + 2.25 = 6.25$$

Ward — size factor, ΔESS , height

$$\frac{|A||B|}{|A|+|B|} = \frac{2 \times 1}{2+1} = \frac{2}{3} = 0.6667$$

$$\Delta ESS(A, B) = 0.6667 \times 6.25 = 4.1667$$

$$d_{\text{ward}}(A, B) = \sqrt{2 \times 4.1667} = \sqrt{8.3333} = \mathbf{2.8868}$$

Final merge: $A' = \{P_1, P_2, P_3\}$ against $B' = \{P_4, P_5\}$

$|A'| = 3$, $|B'| = 2$, so $3 \times 2 = 6$ cross-pairs, all from the Section 1 matrix:

$$d(P_1, P_4) = 5.8310, \quad d(P_1, P_5) = 6.7082, \quad d(P_2, P_4) = 5.3852,$$

$$d(P_2, P_5) = 6.3246, \quad d(P_3, P_4) = 3.1623, \quad d(P_3, P_5) = 4.1231.$$

$$d_{\text{single}} = \min(5.8310, 6.7082, 5.3852, 6.3246, 3.1623, 4.1231) = \mathbf{3.1623}$$

$$d_{\text{complete}} = \max(5.8310, 6.7082, 5.3852, 6.3246, 3.1623, 4.1231) = \mathbf{6.7082}$$

$$\text{sum} = 5.8310 + 6.7082 + 5.3852 + 6.3246 + 3.1623 + 4.1231 = 31.5344$$

$$d_{\text{avg}} = \frac{31.5344}{6} = \mathbf{5.2557}$$

Final merge — Ward

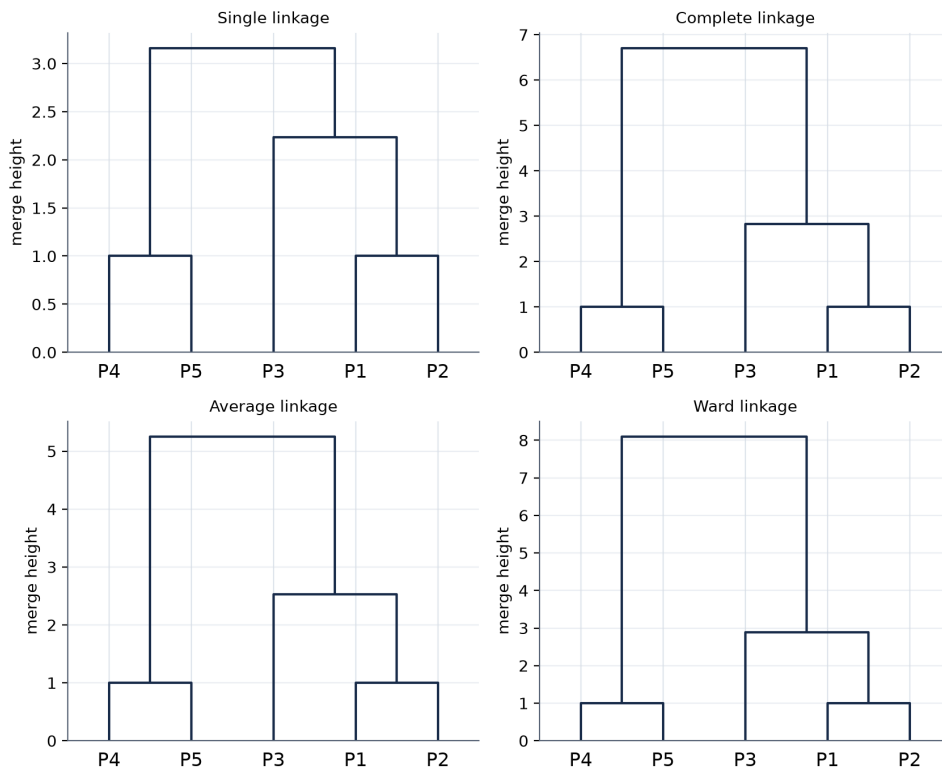
$$\mu_{A'} = \left(\frac{1+1+3}{3}, \frac{1+2+3}{3} \right) = \left(\frac{5}{3}, \frac{6}{3} \right) = (1.6667, 2)$$

$$\mu_{B'} = \left(\frac{6+7}{2}, \frac{4+4}{2} \right) = \left(\frac{13}{2}, \frac{8}{2} \right) = (6.5, 4)$$

$$\|\mu_{A'} - \mu_{B'}\|^2 = (1.6667 - 6.5)^2 + (2 - 4)^2 = (-4.8333)^2 + (-2)^2 = 23.3611 + 4 = 27.3611$$

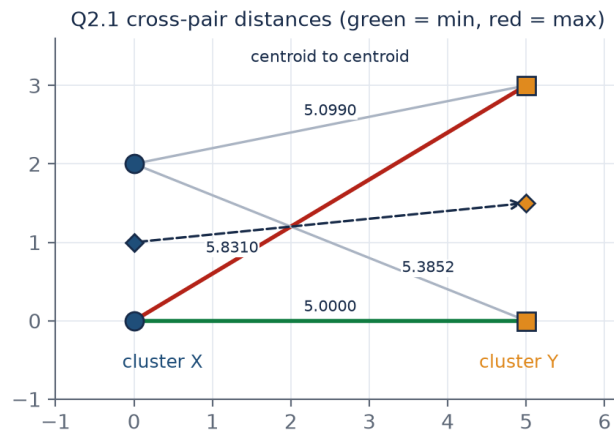
$$\frac{|A'| |B'|}{|A'| + |B'|} = \frac{3 \times 2}{3 + 2} = \frac{6}{5} = 1.2$$

$$\Delta\text{ESS} = 1.2 \times 27.3611 = 32.8333, \quad d_{\text{ward}} = \sqrt{2 \times 32.8333} = \sqrt{65.6667} = \mathbf{8.1035}$$



Questions

Q2.1 $X = \{(0, 0), (0, 2)\}$, $Y = \{(5, 0), (5, 3)\}$. Compute the four cross-pair distances, then d_{single} , d_{complete} , d_{avg} , ΔESS and d_{ward} between X and Y .



Solution.

$$d((0,0), (5,0)) = 5.0000, \quad d((0,0), (5,3)) = \sqrt{34} = 5.8310,$$

$$d((0,2), (5,0)) = \sqrt{29} = 5.3852, \quad d((0,2), (5,3)) = \sqrt{26} = 5.0990.$$

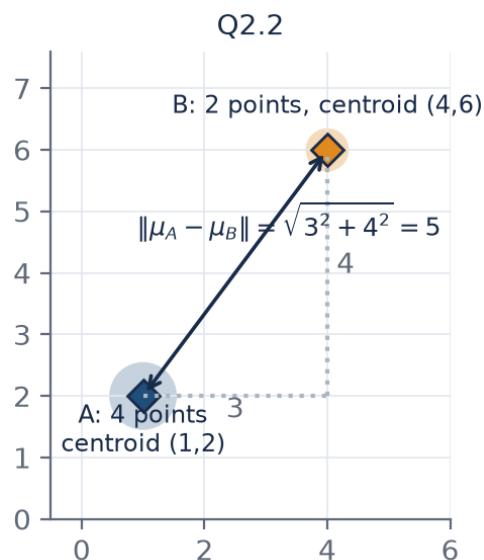
$$d_{\text{single}} = \mathbf{5.0000}, \quad d_{\text{complete}} = \mathbf{5.8310},$$

$$d_{\text{avg}} = \frac{5.0000 + 5.8310 + 5.3852 + 5.0990}{4} = \frac{21.3152}{4} = \mathbf{5.3288}.$$

$$\mu_X = (0,1), \mu_Y = (5,1.5), \|\mu_X - \mu_Y\|^2 = 25 + 0.25 = 25.25.$$

$$\Delta\text{ESS} = \frac{2 \times 2}{4} \times 25.25 = \mathbf{25.2500}, \quad d_{\text{ward}} = \sqrt{50.5} = \mathbf{7.1063}.$$

Q2.2 Cluster A: 4 points, centroid (1,2). Cluster B: 2 points, centroid (4,6). Compute $\Delta\text{ESS}(A,B)$, $d_{\text{ward}}(A,B)$, and the size and centroid of $A \cup B$.



Solution.

$$\|\mu_A - \mu_B\|^2 = (1-4)^2 + (2-6)^2 = 9 + 16 = 25$$

$$\Delta\text{ESS} = \frac{4 \times 2}{4 + 2} \times 25 = \mathbf{33.3333}, \quad d_{\text{ward}} = \sqrt{66.6667} = \mathbf{8.1650}$$

$$|A \cup B| = \mathbf{6}, \quad \mu_{A \cup B} = \frac{4(1, 2) + 2(4, 6)}{6} = \frac{(12, 20)}{6} = (\mathbf{2}, \mathbf{3.3333})$$

Q2.3 Points 0, 1, 3, 7, 8. Run Ward's method to completion. At each step compute d_{ward} for every candidate pair before choosing the smallest. (For two singletons, $d_{\text{ward}} = |a - b|$.)



Solution.

Singletons

$\sqrt{\frac{2 \cdot 1 \cdot 1}{2}} |a - b| = |a - b|$. Smallest: $d(0, 1) = 1$, $d(7, 8) = 1$.

Merges 1, 2: $\{0, 1\}$, $\{7, 8\}$ at **1**

$\mu_{\{0,1\}} = 0.5$, $\mu_{\{7,8\}} = 7.5$.

Merge 3

3 vs $\{0, 1\}$: $\sqrt{\frac{2 \cdot 1 \cdot 2}{3}} \times |3 - 0.5| = 1.1547 \times 2.5 = \mathbf{2.8868}$

3 vs $\{7, 8\}$: $1.1547 \times |3 - 7.5| = 5.1962$

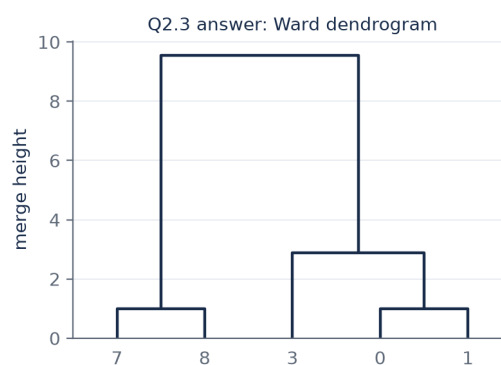
$\{0, 1\}$ vs $\{7, 8\}$: $\sqrt{\frac{2 \cdot 2 \cdot 2}{4}} \times |0.5 - 7.5| = \sqrt{2} \times 7 = 9.8995$

Merge $\{0, 1, 3\}$ at 2.8868; $\mu_{\{0,1,3\}} = \frac{4}{3} = 1.3333$.

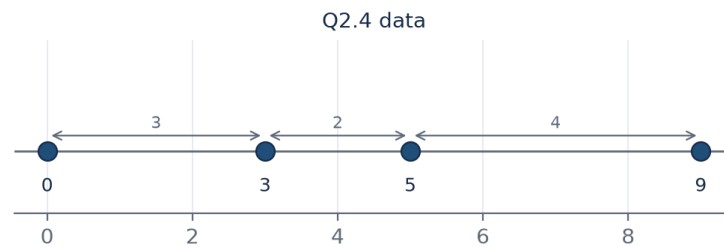
Merge 4

$\sqrt{\frac{2 \cdot 3 \cdot 2}{5}} \times |1.3333 - 7.5| = 1.5492 \times 6.1667 = \mathbf{9.5534}$

$(0, 1)@1$, $(7, 8)@1$, $(3, \{0, 1\})@2.8868$, $(\{0, 1, 3\}, \{7, 8\})@9.5534$



Q2.4 Points 0, 3, 5, 9. **(a)** All six distances. **(b)** Single linkage to completion. **(c)** Complete linkage to completion. **(d)** Average linkage to completion. **(e)** Ratio of the true distance $d(0,9) = 9$ to each linkage's final merge height.



Solution.

(a)

$$d(0,3) = 3, d(0,5) = 5, d(0,9) = 9, d(3,5) = 2, d(3,9) = 6, d(5,9) = 4.$$

(b) Single

$$\{3,5\}@2. \quad d(0,\{3,5\}) = \min(3,5) = 3, \quad d(\{3,5\},9) = \min(6,4) = 4, \quad d(0,9) = 9:$$

$$\{0,3,5\}@3. \quad d(\{0,3,5\},9) = \min(9,6,4) = 4.$$

(c) Complete

$$\{3,5\}@2. \quad d(0,\{3,5\}) = \max(3,5) = 5, \quad d(\{3,5\},9) = \max(6,4) = 6, \quad d(0,9) = 9:$$

$$\{0,3,5\}@5. \quad d(\{0,3,5\},9) = \max(9,6,4) = 9.$$

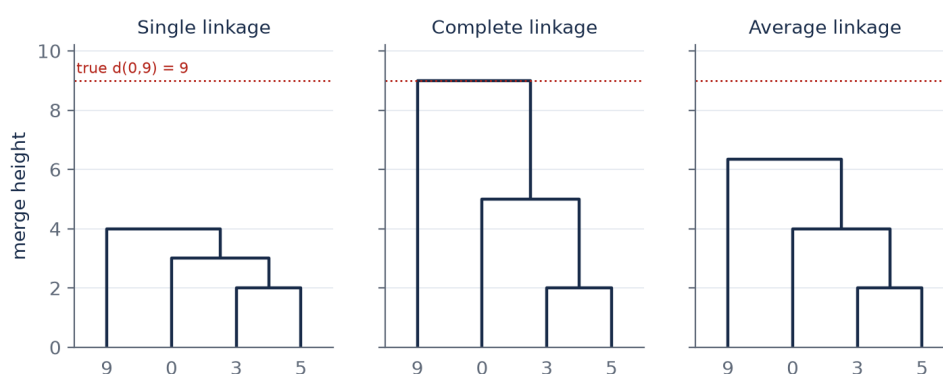
(d) Average

$$\{3,5\}@2. \quad d(0,\{3,5\}) = \frac{3+5}{2} = 4, \quad d(\{3,5\},9) = \frac{6+4}{2} = 5, \quad d(0,9) = 9: \quad \{0,3,5\}@4.$$

$$d(\{0,3,5\},9) = \frac{9+6+4}{3} = 6.3333.$$

(e)

$$\text{Single } 9/4 = \mathbf{2.25}; \text{ average } 9/6.3333 = \mathbf{1.42}; \text{ complete } 9/9 = \mathbf{1.00}.$$



Q2.5 For the points of Q1.1 (1, 2, 4, 8, 13): **(a)** complete linkage to completion; **(b)** average linkage to completion; **(c)** the $k = 2$ partition from each of the single (Q1.1), complete and average dendrograms.

Solution.

(a) Complete

$\{1, 2\}@1$.

$$d(\{1, 2\}, 4) = \max(3, 2) = 3, \quad d(\{1, 2\}, 8) = \max(7, 6) = 7,$$

$$d(\{1, 2\}, 13) = \max(12, 11) = 12, \quad d(4, 8) = 4, \quad d(4, 13) = 9, \quad d(8, 13) = 5.$$

$\{1, 2, 4\}@3$.

$$d(\{1, 2, 4\}, 8) = \max(7, 6, 4) = 7, \quad d(\{1, 2, 4\}, 13) = 11, \quad d(8, 13) = 5.$$

$\{8, 13\}@5$. $d(\{1, 2, 4\}, \{8, 13\}) = \max(7, 6, 4, 12, 11, 9) = \mathbf{12}$.

(b) *Average*

$\{1, 2\}@1$.

$$d(\{1, 2\}, 4) = \frac{3+2}{2} = 2.5, \quad d(\{1, 2\}, 8) = \frac{7+6}{2} = 6.5, \quad d(\{1, 2\}, 13) = \frac{12+11}{2} = 11.5,$$

$$d(4, 8) = 4, \quad d(4, 13) = 9, \quad d(8, 13) = 5.$$

$\{1, 2, 4\}@2.5$.

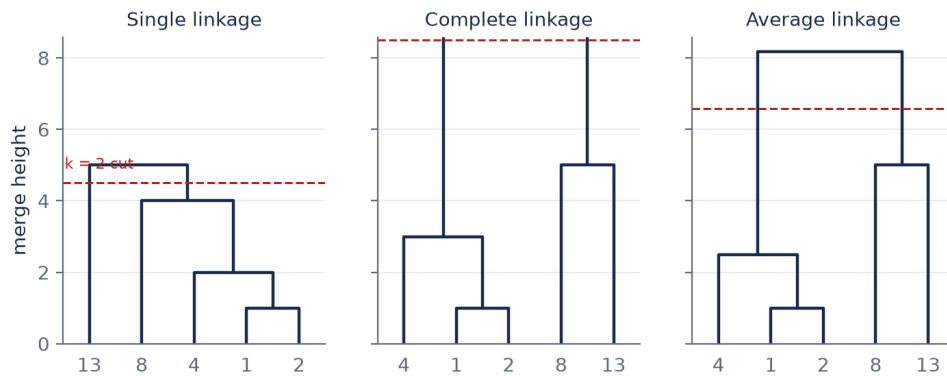
$$d(\{1, 2, 4\}, 8) = \frac{7+6+4}{3} = 5.6667, \quad d(\{1, 2, 4\}, 13) = \frac{12+11+9}{3} = 10.6667, \quad d(8, 13) = 5.$$

$\{8, 13\}@5$.

$$d(\{1, 2, 4\}, \{8, 13\}) = \frac{7+6+4+12+11+9}{6} = \frac{49}{6} = \mathbf{8.1667}.$$

(c) $k = 2$

Single: $\{1, 2, 4, 8\}, \{13\}$. Complete: $\{1, 2, 4\}, \{8, 13\}$. Average: $\{1, 2, 4\}, \{8, 13\}$.



3 Dendrogram interpretation

Worked example 3.1 — cuts, k -intervals and cophenetic distances

Merge table (Section 1):

$(P_1, P_2)@1.0000$, $(P_4, P_5)@1.0000$, $(P_3, \{P_1, P_2\})@2.2361$, $(\{P_1, P_2, P_3\}, \{P_4, P_5\})@3.1623$.

$n = 5$; sorted heights $h_1 = 1.0000$, $h_2 = 1.0000$, $h_3 = 2.2361$, $h_4 = 3.1623$.

Cut at $h = 2.0$

Test each merge: $1.0000 \leq 2.0$ yes; $1.0000 \leq 2.0$ yes; $2.2361 \leq 2.0$ no; $3.1623 \leq 2.0$ no. Merges kept: 2. Clusters = $5 - 2 = 3$. Apply the two kept merges to the five singletons: $\{P_1, P_2\}$, $\{P_3\}$, $\{P_4, P_5\}$.

Cut at $h = 2.8$

$1.0000 \leq 2.8$ yes; $1.0000 \leq 2.8$ yes; $2.2361 \leq 2.8$ yes; $3.1623 \leq 2.8$ no. Merges kept: 3. Clusters = $5 - 3 = 2$: $\{P_1, P_2, P_3\}$, $\{P_4, P_5\}$.

Interval of h giving exactly k clusters: $[h_{n-k}, h_{n-k+1})$

$k = 2$: $n - k = 3$, $[h_3, h_4) = [2.2361, 3.1623)$, width $3.1623 - 2.2361 = 0.9262$

$k = 3$: $n - k = 2$, $[h_2, h_3) = [1.0000, 2.2361)$, width $2.2361 - 1.0000 = 1.2361$

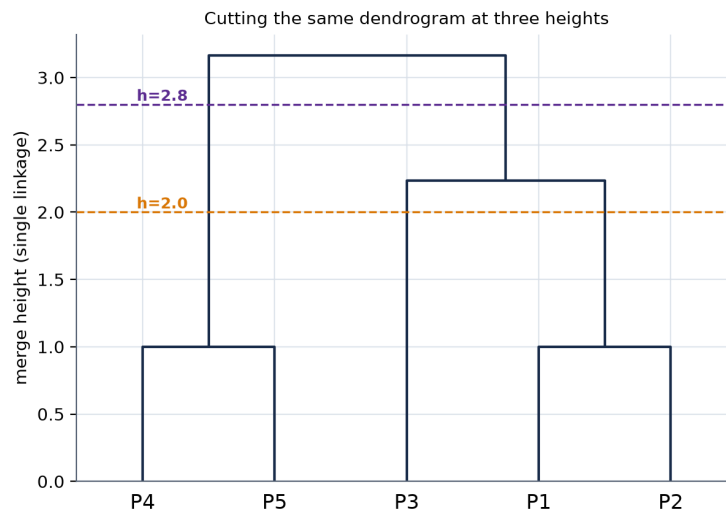
$k = 4$: $n - k = 1$, $[h_1, h_2) = [1.0000, 1.0000)$, width 0 (empty)

Cophenetic distance P_1, P_3

Merge 1 forms $\{P_1, P_2\}$: P_3 not included. Merge 2 forms $\{P_4, P_5\}$: neither. Merge 3 forms $\{P_1, P_2, P_3\}$: both included for the first time. $d_{\text{coph}}(P_1, P_3) = h_3 = 2.2361$. True $d(P_1, P_3) = 2.8284$.

Cophenetic distance P_2, P_5

Merges 1–3 never place P_2 and P_5 together; merge 4 does. $d_{\text{coph}}(P_2, P_5) = h_4 = 3.1623$. True $d(P_2, P_5) = 6.3246$.



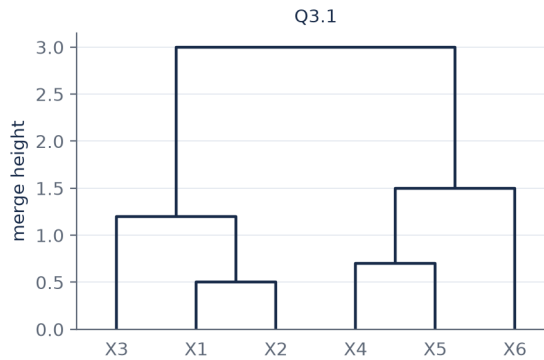
Questions

Q3.1 Six points $X_1, \dots, X_6$; merges in order:

$$(X_1, X_2)@0.5, \quad (X_4, X_5)@0.7, \quad (X_3, \{X_1, X_2\})@1.2,$$

$$(\{X_4, X_5\}, X_6)@1.5, \quad (\{X_1, X_2, X_3\}, \{X_4, X_5, X_6\})@3.0$$

Give the clusters at (a) $h = 1.0$, (b) $h = 1.3$, (c) $h = 2.0$, (d) $h = 3.0$.



Solution.

(a) $h = 1.0$: 2 merges, 4 clusters

$\{X_1, X_2\}, \{X_4, X_5\}, \{X_3\}, \{X_6\}$.

(b) $h = 1.3$: 3 merges, 3 clusters

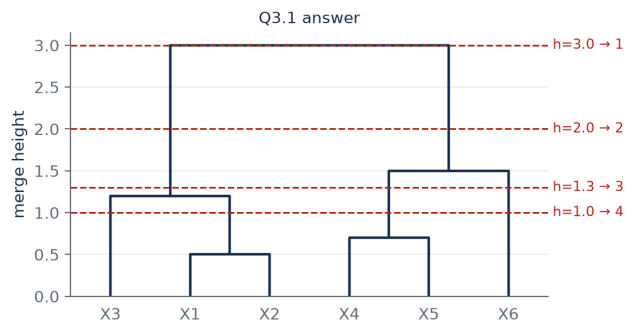
$\{X_1, X_2, X_3\}, \{X_4, X_5\}, \{X_6\}$.

(c) $h = 2.0$: 4 merges, 2 clusters

$\{X_1, X_2, X_3\}, \{X_4, X_5, X_6\}$.

(d) $h = 3.0$: 5 merges, 1 cluster

$\{X_1, \dots, X_6\}$.

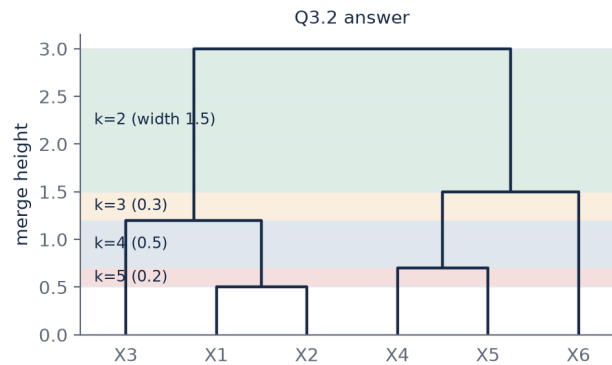


Q3.2 For the merge table of Q3.1, give the interval $[h_{n-k}, h_{n-k+1})$ of cut heights yielding exactly k clusters, and its width, for $k = 2, 3, 4, 5$. Which k has the widest interval?

Solution. $n = 6$; $h_1 = 0.5$, $h_2 = 0.7$, $h_3 = 1.2$, $h_4 = 1.5$, $h_5 = 3.0$.

k	$n - k$	interval	width
2	4	$[h_4, h_5) = [1.5, 3.0)$	1.5
3	3	$[h_3, h_4) = [1.2, 1.5)$	0.3
4	2	$[h_2, h_3) = [0.7, 1.2)$	0.5
5	1	$[h_1, h_2) = [0.5, 0.7)$	0.2

Widest: $k = 2$.

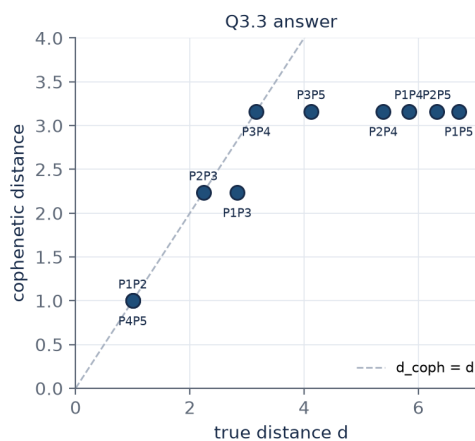


Q3.3 Using the Section 1 merge table, give d_{coph} for all ten pairs of $P_1, \dots, P_5$ alongside the true distance. In how many pairs is $d_{\text{coph}} < d$?

Solution.

pair	d_{coph}	true d	merge
P_1P_2	1.0000	1.0000	1
P_4P_5	1.0000	1.0000	2
P_1P_3	2.2361	2.8284	3
P_2P_3	2.2361	2.2361	3
P_1P_4	3.1623	5.8310	4
P_1P_5	3.1623	6.7082	4
P_2P_4	3.1623	5.3852	4
P_2P_5	3.1623	6.3246	4
P_3P_4	3.1623	3.1623	4
P_3P_5	3.1623	4.1231	4

$d_{\text{coph}} < d$ in **6** pairs; equal in 4.



Q3.4 Points 0, 1, 3, 7. **(a)** Single linkage to completion. **(b)** The six true and six cophenetic distances, in the same pair order. **(c)** The CPCC.



Solution.

(a)

$(0, 1)@1$; $d(3, \{0, 1\}) = \min(3, 2) = 2$, $d(7, \{0, 1\}) = 6$, $d(3, 7) = 4$: $(3, \{0, 1\})@2$;
 $d(7, \{0, 1, 3\}) = \min(7, 6, 4) = 4$: $(7, \{0, 1, 3\})@4$.

(b)

pair	(0, 1)	(0, 3)	(0, 7)	(1, 3)	(1, 7)	(3, 7)
t	1	3	7	2	6	4
c	1	2	4	2	4	4

(c)

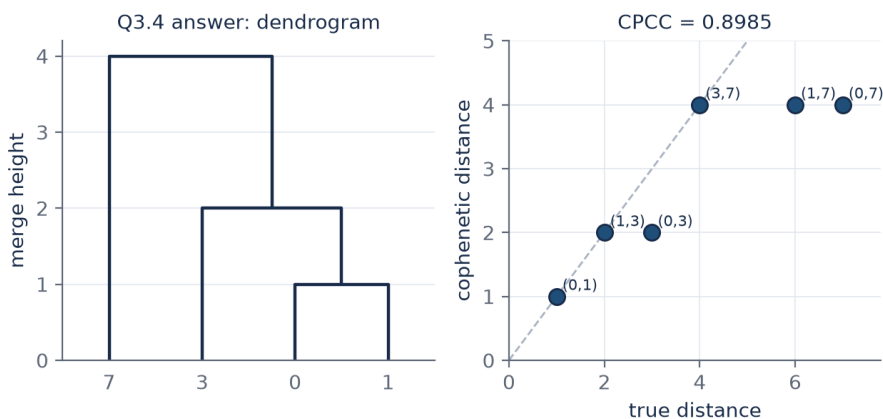
$$\bar{t} = \frac{23}{6} = 3.8333, \bar{c} = \frac{17}{6} = 2.8333.$$

$$\sum (t - \bar{t})(c - \bar{c}) = (-2.8333)(-1.8333) + (-0.8333)(-0.8333) + (3.1667)(1.1667)$$

$$+ (-1.8333)(-0.8333) + (2.1667)(1.1667) + (0.1667)(1.1667) = 13.8333$$

$$\sum (t - \bar{t})^2 = 26.8333, \quad \sum (c - \bar{c})^2 = 8.8333$$

$$\text{CPCC} = \frac{13.8333}{\sqrt{26.8333 \times 8.8333}} = \frac{13.8333}{15.3958} = \mathbf{0.8985}$$

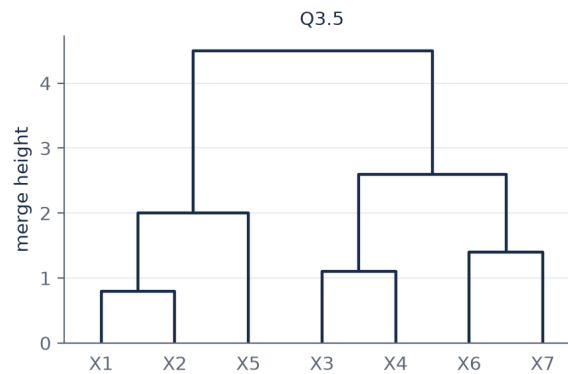


Q3.5 Seven points; merges in order:

$$(X_1, X_2)@0.8, \quad (X_3, X_4)@1.1, \quad (X_6, X_7)@1.4,$$

$$(\{X_1, X_2\}, X_5)@2.0, \quad (\{X_3, X_4\}, \{X_6, X_7\})@2.6, \quad (\text{all})@4.5$$

(a) Number of clusters at $h = 1.2, 2.5, 3.0$. **(b)** At each cut, the points in the same cluster as X_3 . **(c)** $d_{\text{coph}}(X_3, X_6)$ and $d_{\text{coph}}(X_3, X_5)$.



Solution.

(a)

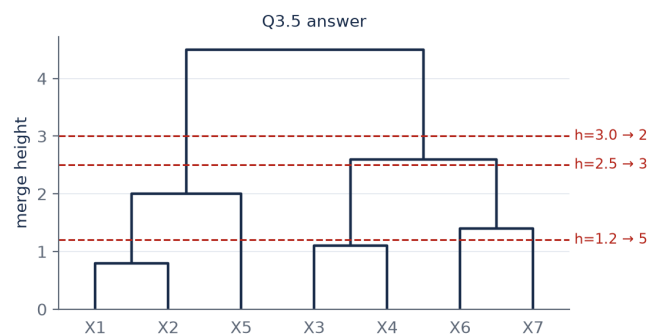
$h = 1.2$: $7 - 2 = \mathbf{5}$. $h = 2.5$: $7 - 4 = \mathbf{3}$. $h = 3.0$: $7 - 5 = \mathbf{2}$.

(b)

$h = 1.2$: X_4 . $h = 2.5$: X_4 . $h = 3.0$: X_4, X_6, X_7 .

(c)

$d_{\text{coph}}(X_3, X_6) = \mathbf{2.6}$; $d_{\text{coph}}(X_3, X_5) = \mathbf{4.5}$.



4 DBSCAN: ε -neighbourhood and MinPts

Worked example 4.1 — *classifying eight points*

$A(1, 1)$, $B(1, 2)$, $C(2, 1)$, $D(2, 2)$, $H(2, 3.2)$, $E(5, 5)$, $F(5.5, 5.3)$, $I(9, 9)$; $\varepsilon = 1.5$, $\text{MINPTS} = 4$.

Distances among A, B, C, D, H

$$d(A, B) = \sqrt{(1-1)^2 + (2-1)^2} = \sqrt{0+1} = 1.0000$$

$$d(A, C) = \sqrt{(2-1)^2 + (1-1)^2} = \sqrt{1+0} = 1.0000$$

$$d(A, D) = \sqrt{(2-1)^2 + (2-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$$

$$d(B, C) = \sqrt{(2-1)^2 + (1-2)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$$

$$d(B, D) = \sqrt{(2-1)^2 + (2-2)^2} = \sqrt{1+0} = 1.0000$$

$$d(C, D) = \sqrt{(2-2)^2 + (2-1)^2} = \sqrt{0+1} = 1.0000$$

$$d(H, A) = \sqrt{(2-1)^2 + (3.2-1)^2} = \sqrt{1+4.84} = \sqrt{5.84} = 2.4166$$

$$d(H, B) = \sqrt{(2-1)^2 + (3.2-2)^2} = \sqrt{1+1.44} = \sqrt{2.44} = 1.5620$$

$$d(H, C) = \sqrt{(2-2)^2 + (3.2-1)^2} = \sqrt{0+4.84} = \sqrt{4.84} = 2.2000$$

$$d(H, D) = \sqrt{(2-2)^2 + (3.2-2)^2} = \sqrt{0+1.44} = \sqrt{1.44} = 1.2000$$

Distances involving E, F, I

$$d(E, F) = \sqrt{(5.5-5)^2 + (5.3-5)^2} = \sqrt{0.25+0.09} = \sqrt{0.34} = 0.5831$$

$$d(D, E) = \sqrt{(5-2)^2 + (5-2)^2} = \sqrt{9+9} = \sqrt{18} = 4.2426$$

$$d(H, E) = \sqrt{(5-2)^2 + (5-3.2)^2} = \sqrt{9+3.24} = \sqrt{12.24} = 3.4986$$

$$d(D, F) = \sqrt{(5.5-2)^2 + (5.3-2)^2} = \sqrt{12.25+10.89} = \sqrt{23.14} = 4.8104$$

$$d(H, F) = \sqrt{(5.5-2)^2 + (5.3-3.2)^2} = \sqrt{12.25+4.41} = \sqrt{16.66} = 4.0817$$

$$d(F, I) = \sqrt{(9-5.5)^2 + (9-5.3)^2} = \sqrt{12.25+13.69} = \sqrt{25.94} = 5.0931$$

$$d(E, I) = \sqrt{(9-5)^2 + (9-5)^2} = \sqrt{16+16} = \sqrt{32} = 5.6569$$

All other pairs (A, B, C to E, F, I ; D to I ; H to I) are farther than these and therefore $> \varepsilon$.

Full distance table

	B	C	D	H	E	F	I
A	1.0000	1.0000	1.4142	2.4166	5.6569	6.2241	11.3137
B		1.4142	1.0000	1.5620	5.0000	5.5803	10.6301
C			1.0000	2.2000	5.0000	5.5444	10.6301
D				1.2000	4.2426	4.8104	9.8995
H					3.4986	4.0817	9.0907
E						0.5831	5.6569
F							5.0931

Neighbourhoods (entries ≤ 1.5 in each point's row and column, plus the point itself)

$$N_\varepsilon(A) = \{A, B, C, D\} \text{ (1.0000, 1.0000, 1.4142 } \leq 1.5; 2.4166 > 1.5) \quad |N| = 4$$

$$N_\varepsilon(B) = \{B, A, C, D\} \text{ (1.0000, 1.4142, 1.0000 } \leq 1.5; 1.5620 > 1.5) \quad |N| = 4$$

$$N_\varepsilon(C) = \{C, A, B, D\} \text{ (1.0000, 1.4142, 1.0000 } \leq 1.5; 2.2000 > 1.5) \quad |N| = 4$$

$$N_\varepsilon(D) = \{D, A, B, C, H\} \text{ (1.4142, 1.0000, 1.0000, 1.2000 } \leq 1.5) \quad |N| = 5$$

$$N_\varepsilon(H) = \{H, D\} \text{ (1.2000 } \leq 1.5; 1.5620, 2.2000, 2.4166 > 1.5) \quad |N| = 2$$

$$N_\varepsilon(E) = \{E, F\} \text{ (0.5831 } \leq 1.5; 3.4986, 4.2426, \dots > 1.5) \quad |N| = 2$$

$$N_\varepsilon(F) = \{F, E\} \text{ (0.5831 } \leq 1.5) \quad |N| = 2 \quad N_\varepsilon(I) = \{I\} \text{ (smallest 5.0931 } > 1.5) \quad |N| = 1$$

Core test: $|N_\varepsilon(p)| \geq 4$?

A: $4 \geq 4$ core. B: $4 \geq 4$ core. C: $4 \geq 4$ core. D: $5 \geq 4$ core. H: $2 < 4$ not core. E: $2 < 4$ not core. F: $2 < 4$ not core. I: $1 < 4$ not core.

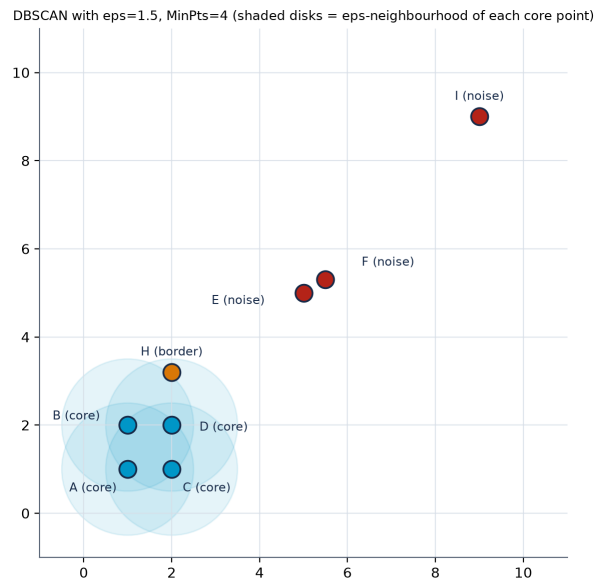
Border test for the non-core points: within 1.5 of a core point?

H: $d(H, D) = 1.2000 \leq 1.5$, D core $\Rightarrow$ **border**. E: nearest core is D, $d(E, D) = 4.2426 > 1.5 \Rightarrow$ **noise**. F: nearest core is D, $d(F, D) = 4.8104 > 1.5 \Rightarrow$ **noise**. I: nearest core is D, $d(I, D) = 9.8995 > 1.5 \Rightarrow$ **noise**.

p	$N_\varepsilon(p)$	$ N_\varepsilon(p) $	label
A	$\{A, B, C, D\}$	4	core
B	$\{A, B, C, D\}$	4	core
C	$\{A, B, C, D\}$	4	core
D	$\{A, B, C, D, H\}$	5	core
H	$\{D, H\}$	2	border
E	$\{E, F\}$	2	noise
F	$\{E, F\}$	2	noise
I	$\{I\}$	1	noise

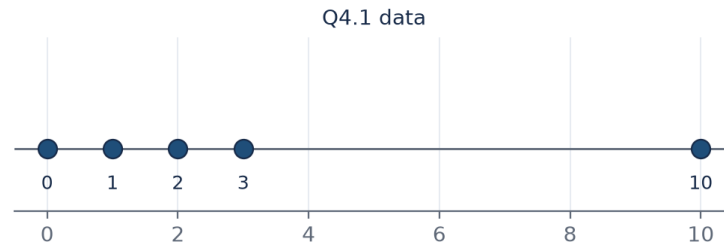
Clusters

Core points A, B, C, D are pairwise within 1.5 (largest pair 1.4142): one chain. H attaches via D. Cluster $\{A, B, C, D, H\}$; noise $\{E, F, I\}$; noise fraction $3/8 = 0.375 = 37.5\%$.



Questions

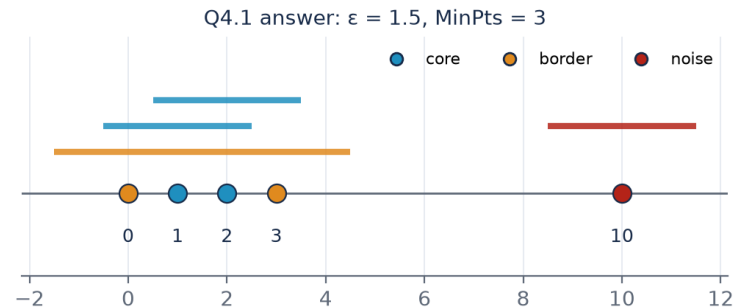
Q4.1 Points 0, 1, 2, 3, 10; $\varepsilon = 1.5$, MINPTS = 3. Tabulate $N_\varepsilon(p)$ and $|N_\varepsilon(p)|$ and classify each point.



Solution.

p	$N_\varepsilon(p)$	size	label
0	{0, 1}	2	border (1 core)
1	{0, 1, 2}	3	core
2	{1, 2, 3}	3	core
3	{2, 3}	2	border (2 core)
10	{10}	1	noise

Cluster {0, 1, 2, 3}; noise {10}.



Q4.2 Same points, MINPTS = 3. For $\varepsilon = 0.5, 1.0, 2.0$: counts of core, border and noise points, and number of clusters.

Solution.

$\varepsilon = 0.5$

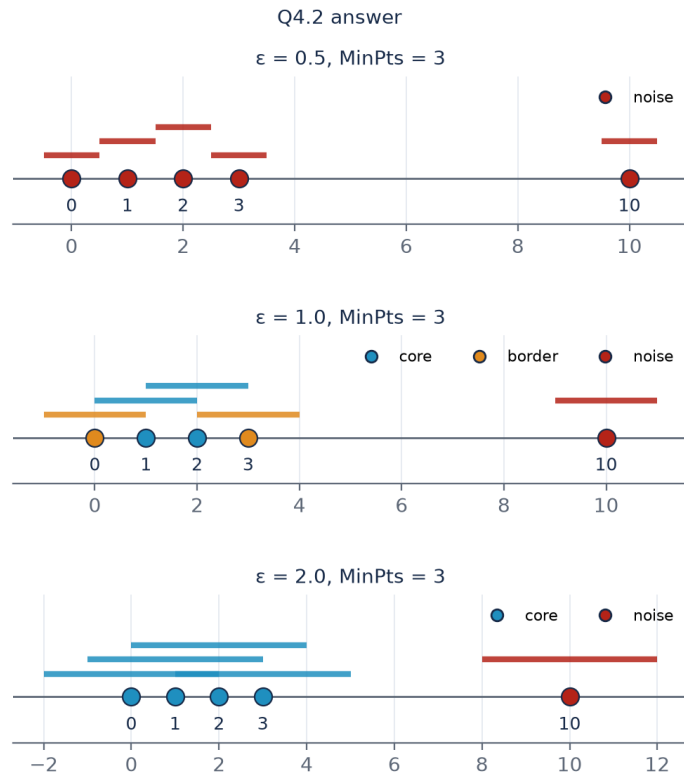
Sizes 1, 1, 1, 1, 1. Core 0, border 0, noise 5; **0 clusters**.

$\varepsilon = 1.0$

Sizes 2, 3, 3, 2, 1. Core {1, 2}, border {0, 3}, noise {10}: (2, 2, 1); **1 cluster**.

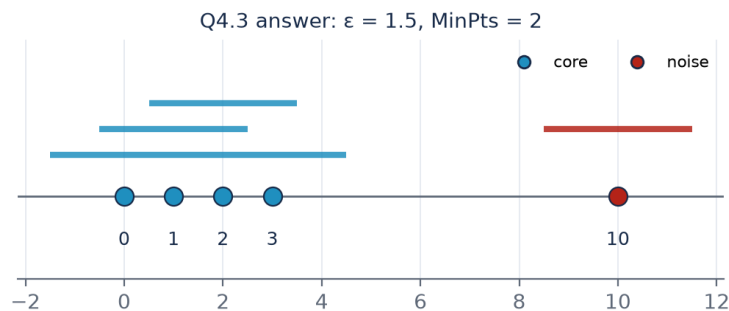
$\varepsilon = 2.0$

$N(0) = \{0, 1, 2\}$, $N(1) = \{0, 1, 2, 3\}$, $N(2) = \{0, 1, 2, 3\}$, $N(3) = \{1, 2, 3\}$, $N(10) = \{10\}$; sizes 3, 4, 4, 3, 1. Core {0, 1, 2, 3}, border none, noise {10}: (4, 0, 1); **1 cluster**.



Q4.3 Same points, $\epsilon = 1.5$, $\text{MINPTS} = 2$. Reclassify and state which points changed label.

Solution. Sizes 2, 3, 3, 2, 1. Core $\{0, 1, 2, 3\}$; noise $\{10\}$. Changed: 0 and 3, border $\rightarrow$ core.



Q4.4 Points 0, 1.2, 2.0, 3.5, 5.0, 9.0; $\text{MINPTS} = 3$. **(a)** For each point, its distances to the other five in increasing order and its 2-distance. **(b)** The six 2-distances sorted decreasing. **(c)** Number of core points at $\epsilon = 1.5, 2.0, 3.6$; verify by classifying all points at $\epsilon = 1.5$.



Solution.

(a)

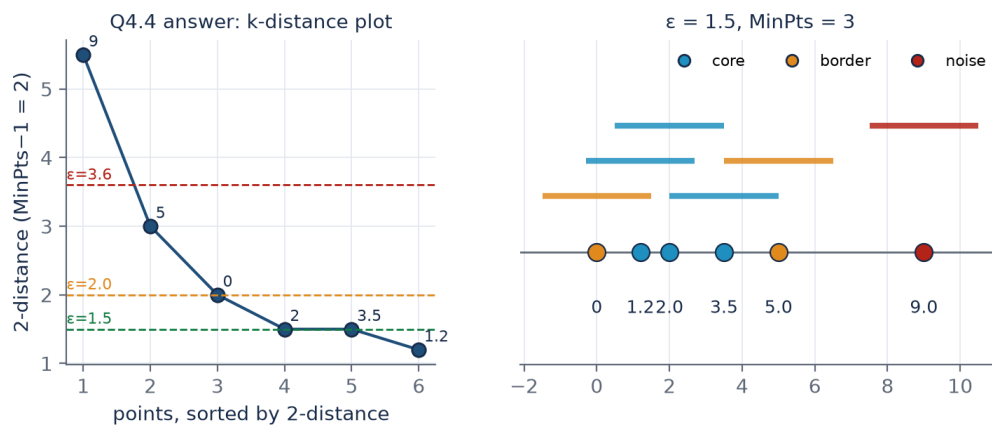
p	sorted distances	2-distance
0	1.2, 2.0, 3.5, 5.0, 9.0	2.0
1.2	0.8, 1.2, 2.3, 3.8, 7.8	1.2
2.0	0.8, 1.5, 2.0, 3.0, 7.0	1.5
3.5	1.5, 1.5, 2.3, 3.5, 5.5	1.5
5.0	1.5, 3.0, 3.8, 4.0, 5.0	3.0
9.0	4.0, 5.5, 7.0, 7.8, 9.0	5.5

(b)

5.5, 3.0, 2.0, 1.5, 1.5, 1.2.

(c)

$\varepsilon = 1.5$: **3** core; $\varepsilon = 2.0$: **4**; $\varepsilon = 3.6$: **5**. At $\varepsilon = 1.5$: $N(0) = \{0, 1.2\}$ (2); $N(1.2) = \{0, 1.2, 2.0\}$ (3); $N(2.0) = \{1.2, 2.0, 3.5\}$ (3); $N(3.5) = \{2.0, 3.5, 5.0\}$ (3); $N(5.0) = \{3.5, 5.0\}$ (2); $N(9.0) = \{9.0\}$ (1). Core 1.2, 2.0, 3.5; border 0, 5.0; noise 9.0.



5 Noise and border points: forming the clusters

Worked example 5.1 — *two clusters, a shared border point, one noise point*

$a(0,0)$, $b(1,0)$, $c(0,1)$, $d(1,1)$, $e(2,2)$, $f(3,3)$, $g(4,3)$, $h(3,4)$, $i(4,4)$, $j(7,0)$; $\varepsilon = 1.5$, $\text{MINPTS} = 4$.

Distances inside the first square a, b, c, d

$$\begin{aligned} d(a,b) &= \sqrt{1^2 + 0^2} = 1.0000, & d(a,c) &= \sqrt{0^2 + 1^2} = 1.0000, & d(a,d) &= \sqrt{1^2 + 1^2} = 1.4142, \\ d(b,c) &= \sqrt{(-1)^2 + 1^2} = 1.4142, & d(b,d) &= \sqrt{0^2 + 1^2} = 1.0000, & d(c,d) &= \sqrt{1^2 + 0^2} = 1.0000. \end{aligned}$$

Distances inside the second square f, g, h, i

$$\begin{aligned} d(f,g) &= \sqrt{1^2 + 0^2} = 1.0000, & d(f,h) &= \sqrt{0^2 + 1^2} = 1.0000, & d(f,i) &= \sqrt{1^2 + 1^2} = 1.4142, \\ d(g,h) &= \sqrt{(-1)^2 + 1^2} = 1.4142, & d(g,i) &= \sqrt{0^2 + 1^2} = 1.0000, & d(h,i) &= \sqrt{1^2 + 0^2} = 1.0000. \end{aligned}$$

Distances from $e(2,2)$

$$\begin{aligned} d(e,d) &= \sqrt{(2-1)^2 + (2-1)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142 \\ d(e,f) &= \sqrt{(3-2)^2 + (3-2)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142 \\ d(e,b) &= \sqrt{(2-1)^2 + (2-0)^2} = \sqrt{1+4} = 2.2361, & d(e,c) &= \sqrt{4+1} = 2.2361, \\ d(e,g) &= \sqrt{(4-2)^2 + (3-2)^2} = \sqrt{4+1} = 2.2361, & d(e,h) &= \sqrt{1+4} = 2.2361, \\ d(e,a) &= \sqrt{4+4} = 2.8284, & d(e,i) &= \sqrt{4+4} = 2.8284. \end{aligned}$$

Between the squares, and from $j(7,0)$

$$\begin{aligned} d(d,f) &= \sqrt{(3-1)^2 + (3-1)^2} = \sqrt{8} = 2.8284 \text{ (closest cross pair)} \\ d(j,g) &= \sqrt{(7-4)^2 + (0-3)^2} = \sqrt{9+9} = 4.2426 \text{ (closest to } j) \end{aligned}$$

Full distance table

	b	c	d	e	f	g	h	i	j
a	1.0000	1.0000	1.4142	2.8284	4.2426	5.0000	5.0000	5.6569	7.0000
b		1.4142	1.0000	2.2361	3.6056	4.2426	4.4721	5.0000	6.0000
c			1.0000	2.2361	3.6056	4.4721	4.2426	5.0000	7.0711
d				1.4142	2.8284	3.6056	3.6056	4.2426	6.0828
e					1.4142	2.2361	2.2361	2.8284	5.3852
f						1.0000	1.0000	1.4142	5.0000
g							1.4142	1.0000	4.2426
h								1.0000	5.6569
i									5.0000

Neighbourhoods ($\text{entries} \leq 1.5$) and core test (≥ 4)

$$\begin{aligned}
 N(a) &= \{a, b, c, d\}, |N| = 4 \geq 4 \text{ core} & N(b) &= \{b, a, c, d\}, |N| = 4 \geq 4 \text{ core} \\
 N(c) &= \{c, a, b, d\}, |N| = 4 \geq 4 \text{ core} & N(d) &= \{d, a, b, c, e\}, |N| = 5 \geq 4 \text{ core} \\
 N(e) &= \{e, d, f\}, |N| = 3 < 4 \text{ not core} & N(f) &= \{f, e, g, h, i\}, |N| = 5 \geq 4 \text{ core} \\
 N(g) &= \{g, f, h, i\}, |N| = 4 \geq 4 \text{ core} & N(h) &= \{h, f, g, i\}, |N| = 4 \geq 4 \text{ core} \\
 N(i) &= \{i, f, g, h\}, |N| = 4 \geq 4 \text{ core} & N(j) &= \{j\}, |N| = 1 < 4 \text{ not core}
 \end{aligned}$$

Border test

e : $d(e, d) = 1.4142 \leq 1.5$ with d core, and $d(e, f) = 1.4142 \leq 1.5$ with f core $\Rightarrow$ **border**
 (reachable from two cores). j : nearest point g at $4.2426 > 1.5 \Rightarrow$ **noise**.

p	$N_\epsilon(p)$	size	label
a, b, c	$\{a, b, c, d\}$	4	core
d	$\{a, b, c, d, e\}$	5	core
e	$\{d, e, f\}$	3	border
f	$\{e, f, g, h, i\}$	5	core
g, h, i	$\{f, g, h, i\}$	4	core
j	$\{j\}$	1	noise

Chaining the core points

a, b, c, d : every pair $\leq 1.4142 \leq 1.5 \Rightarrow$ one chain C_1 . f, g, h, i : every pair $\leq 1.4142 \leq 1.5 \Rightarrow$ one chain C_2 . Between chains, the smallest core-to-core distance is $d(d, f) = 2.8284 > 1.5 \Rightarrow C_1$ and C_2 stay separate.

Attaching the border point

e is reachable from $d \in C_1$ and $f \in C_2$; it is assigned to the cluster expanded first (scikit-learn, index order: C_1).

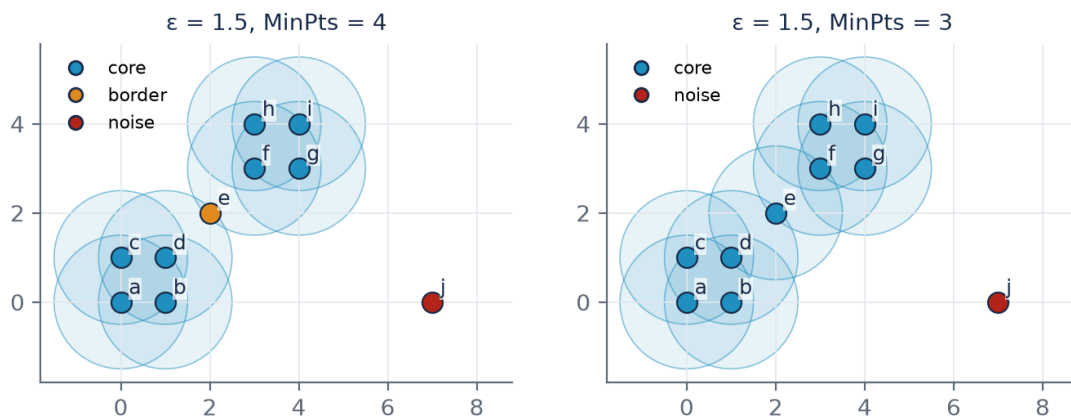
Result, $\text{MINPTS} = 4$

$C_1 = \{a, b, c, d, e\}$, $C_2 = \{f, g, h, i\}$, noise $\{j\}$. Noise fraction $1/10 = 0.10 = 10\%$.

Repeat with $\text{MINPTS} = 3$

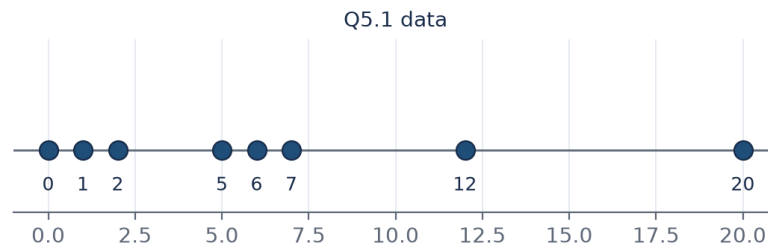
Neighbourhood sizes unchanged: 4, 4, 4, 5, 3, 5, 4, 4, 4, 1. Core test ≥ 3 : $a, b, c, d, e, f, g, h, i$ all core; j ($1 < 3$) not core, and still $4.2426 > 1.5$ from everything $\Rightarrow$ noise. Chaining: $d(d, e) = 1.4142 \leq 1.5$ and $d(e, f) = 1.4142 \leq 1.5$, so C_1-e-C_2 is one chain. **One cluster** $\{a, b, c, d, e, f, g, h, i\}$; noise $\{j\}$.

Worked example 5.1

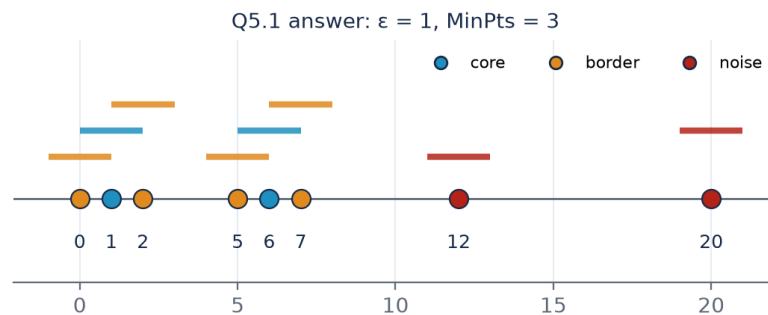


Questions

Q5.1 Points 0, 1, 2, 5, 6, 7, 12, 20; $\varepsilon = 1$, $\text{MinPts} = 3$. Classify every point, list the clusters, and give the noise fraction.



Solution. $N(0) = \{0, 1\}$, $N(1) = \{0, 1, 2\}$, $N(2) = \{1, 2\}$, $N(5) = \{5, 6\}$, $N(6) = \{5, 6, 7\}$, $N(7) = \{6, 7\}$, $N(12) = \{12\}$, $N(20) = \{20\}$; sizes 2, 3, 2, 2, 3, 2, 1, 1. Core $\{1, 6\}$; border $\{0, 2, 5, 7\}$; noise $\{12, 20\}$. $d(1, 6) = 5 > 1$: two clusters $\{0, 1, 2\}$, $\{5, 6, 7\}$. Noise fraction $2/8 = 25\%$.



Q5.2 Same points. (a) $\varepsilon = 1$, $\text{MinPts} = 2$; (b) $\varepsilon = 5$, $\text{MinPts} = 3$. Classify every point and list the clusters.

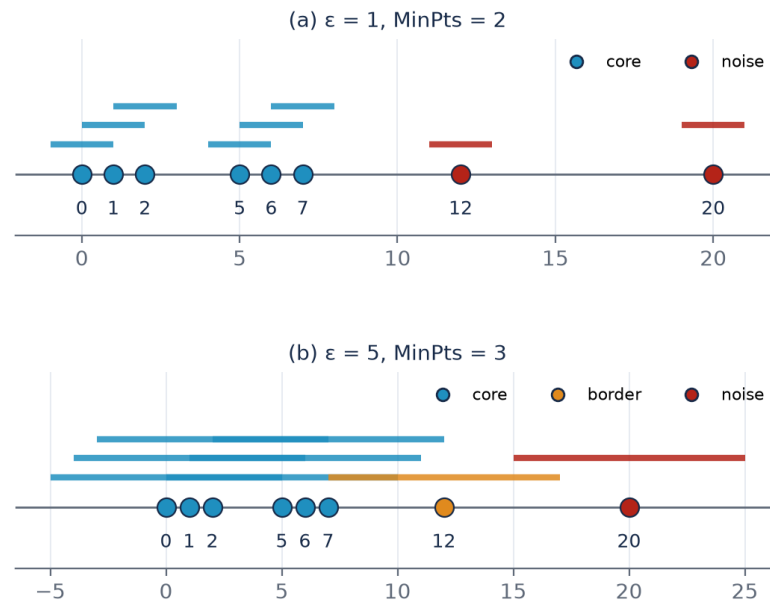
Solution.

(a)

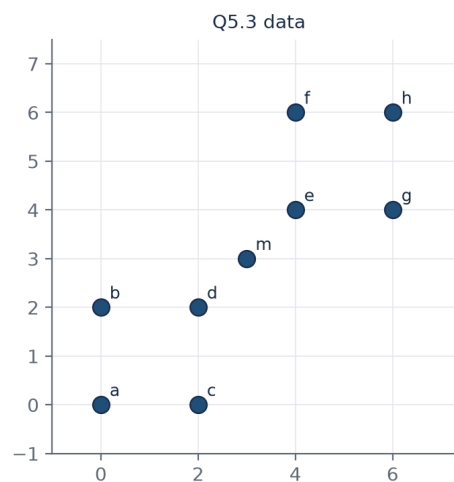
Sizes 2, 3, 2, 2, 3, 2, 1, 1; threshold 2. Core $\{0, 1, 2, 5, 6, 7\}$; noise $\{12, 20\}$. Clusters $\{0, 1, 2\}$, $\{5, 6, 7\}$.

(b)

$N(0) = \{0, 1, 2, 5\}$ (4), $N(1) = \{0, 1, 2, 5, 6\}$ (5), $N(2) = \{0, 1, 2, 5, 6, 7\}$ (6), $N(5) = \{0, 1, 2, 5, 6, 7\}$ (6), $N(6) = \{1, 2, 5, 6, 7\}$ (5), $N(7) = \{2, 5, 6, 7, 12\}$ (5), $N(12) = \{7, 12\}$ (2), $N(20) = \{20\}$ (1). Core $\{0, 1, 2, 5, 6, 7\}$; border $\{12\}$; noise $\{20\}$. $d(2, 5) = 3 \leq 5$: **one cluster** $\{0, 1, 2, 5, 6, 7, 12\}$; noise fraction $1/8 = 12.5\%$.



Q5.3 $a(0,0)$, $b(0,2)$, $c(2,0)$, $d(2,2)$, $m(3,3)$, $e(4,4)$, $f(4,6)$, $g(6,4)$, $h(6,6)$; $\varepsilon = 2$. (a) $|N_\varepsilon(p)|$ for every point. (b) Classification and clusters for MINPTS = 3. (c) The same for MINPTS = 4. (d) Points that changed label between (b) and (c), and the possible clusters for m in (c).



Solution.

(a)

Square side $2 \leq \varepsilon$, diagonal $2.8284 > \varepsilon$. $d(d,m) = d(m,e) = 1.4142$; $d(m,b) = d(m,c) = d(m,f) = d(m,g) = 3.1623$; $d(d,e) = 2.8284$.

p	$N_\varepsilon(p)$	size
a	$\{a, b, c\}$	3
b	$\{a, b, d\}$	3
c	$\{a, c, d\}$	3
d	$\{b, c, d, m\}$	4
m	$\{d, m, e\}$	3
e	$\{m, e, f, g\}$	4
f	$\{e, f, h\}$	3
g	$\{e, g, h\}$	3
h	$\{f, g, h\}$	3

(b) $\text{MINPTS} = 3$

All nine core. d - m - e chain: **one cluster** of 9; no noise.

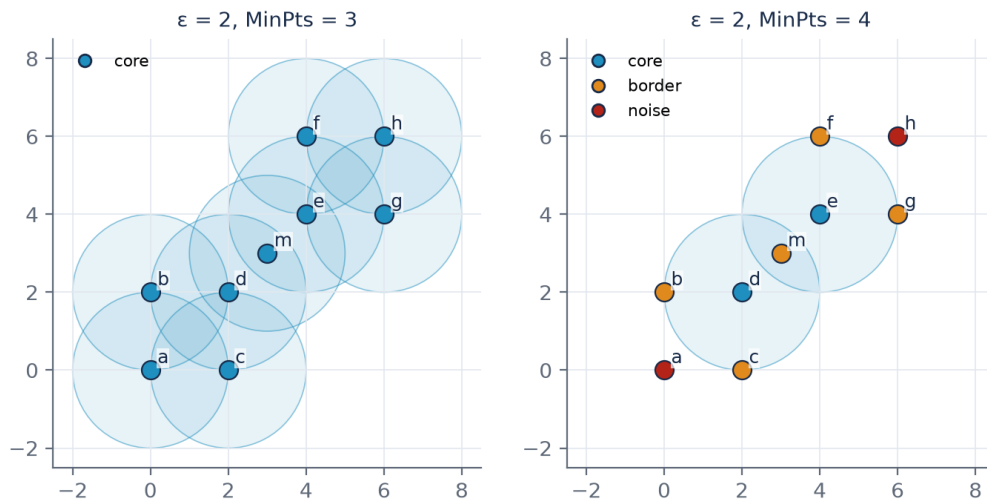
(c) $\text{MINPTS} = 4$

Core $\{d, e\}$; $d(d, e) = 2.8284 > 2$, separate. Border $\{b, c\}$ (via d), $\{f, g\}$ (via e), m (via d and e). Noise $\{a, h\}$. **Two clusters**: $\{b, c, d\}$ and $\{e, f, g\}$, with m in one of them; noise fraction $2/9 = 22.2\%$.

(d)

a, h : core $\rightarrow$ noise. b, c, f, g, m : core $\rightarrow$ border. m can join $\{b, c, d\}$ or $\{e, f, g\}$; sizes (4, 3) or (3, 4).

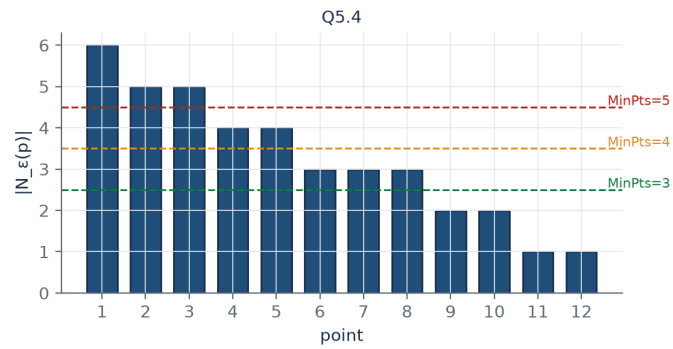
Q5.3 answer



Q5.4 Twelve points with $|N_\varepsilon(p)|$:

6, 5, 5, 4, 4, 3, 3, 3, 2, 2, 1, 1

Every point with size ≥ 2 is within ε of a point of size ≥ 5 . (a) Core count for $\text{MINPTS} = 3, 4, 5$. (b) Noise count in each case. (c) Border count in each case.



Solution.

(a)

MINPTS = 3: **8**. MINPTS = 4: **5**. MINPTS = 5: **3**.

(b)

Size-1 points: **2** noise in every case; all other non-core points are within ϵ of a size- ≥ 5 core point.

(c) $12 - \text{core} - 2$

MINPTS = 3: **2**; 4: **5**; 5: **7**.

6 k-Means vs. hierarchical clustering vs. DBSCAN

Worked example 6.1 — *one dataset, three algorithms*

Points 0, 1, 2, 5, 6, 7, 15.

k-Means, $k = 2$, initial centroids $c_1 = 0$, $c_2 = 15$ — assignment

x	$ x - c_1 = x - 0 $	$ x - c_2 = x - 15 $	assigned to
0	0	15	c_1
1	1	14	c_1
2	2	13	c_1
5	5	10	c_1
6	6	9	c_1
7	7	8	c_1
15	15	0	c_2

Update

$$c_1 = \frac{0 + 1 + 2 + 5 + 6 + 7}{6} = \frac{21}{6} = 3.5, \quad c_2 = \frac{15}{1} = 15$$

Reassignment

x	$ x - 3.5 $	$ x - 15 $	assigned to
0	3.5	15	c_1
1	2.5	14	c_1
2	1.5	13	c_1
5	1.5	10	c_1
6	2.5	9	c_1
7	3.5	8	c_1
15	11.5	0	c_2

No point changed cluster $\Rightarrow$ converged. Centroids 3.5 and 15.

Inertia

$$\begin{aligned} & (0 - 3.5)^2 + (1 - 3.5)^2 + (2 - 3.5)^2 + (5 - 3.5)^2 + (6 - 3.5)^2 + (7 - 3.5)^2 + (15 - 15)^2 \\ &= 12.25 + 6.25 + 2.25 + 2.25 + 6.25 + 12.25 + 0 = \mathbf{41.5} \end{aligned}$$

Partition $\{0, 1, 2, 5, 6, 7\}$, $\{15\}$.

Single linkage — distance matrix ($|x_i - x_j|$)

	1	2	5	6	7	15
0	1	2	5	6	7	15
1		1	4	5	6	14
2			3	4	5	13
5				1	2	10
6					1	9
7						8

Single linkage — merges

Smallest entry 1 (ties: (0, 1), (1, 2), (5, 6), (6, 7)). Merge $\{0, 1\}@1$; $d(\{0, 1\}, 2) = \min(2, 1) = 1$. Merge $\{0, 1, 2\}@1$; $d(\{0, 1, 2\}, 5) = \min(5, 4, 3) = 3$, $d(\{0, 1, 2\}, 6) = \min(6, 5, 4) = 4$, $d(\{0, 1, 2\}, 7) = \min(7, 6, 5) = 5$, $d(\{0, 1, 2\}, 15) = \min(15, 14, 13) = 13$. Merge $\{5, 6\}@1$; $d(\{5, 6\}, 7) = \min(2, 1) = 1$. Merge $\{5, 6, 7\}@1$; $d(\{0, 1, 2\}, \{5, 6, 7\}) = \min(3, 4, 5) = 3$, $d(\{5, 6, 7\}, 15) = \min(10, 9, 8) = 8$. Remaining entries: 3, 13, 8. Merge $\{0, 1, 2, 5, 6, 7\}@3$; $d(\{0, \dots, 7\}, 15) = \min(13, 8) = 8$. Merge all @8.

(0, 1)@1, (2, {0, 1})@1, (5, 6)@1, (7, {5, 6})@1, ({0, 1, 2}, {5, 6, 7})@3, (15, rest)@8

Single linkage — cuts

$k = 2$: undo the last merge (8): $\{0, 1, 2, 5, 6, 7\}, \{15\}$. $k = 3$: undo the last two (8, 3): $\{0, 1, 2\}, \{5, 6, 7\}, \{15\}$.

DBSCAN, $\varepsilon = 1$, $\text{MINPTS} = 3$ — neighbourhoods

$$N(0) = \{0, 1\} (|0 - 1| = 1 \leq 1; |0 - 2| = 2 > 1) \quad |N| = 2 < 3$$

$$N(1) = \{0, 1, 2\} (1, 1 \leq 1) \quad |N| = 3 \geq 3 \text{ core}$$

$$N(2) = \{1, 2\} (|2 - 1| = 1; |2 - 5| = 3 > 1) \quad |N| = 2 < 3$$

$$N(5) = \{5, 6\} (|5 - 6| = 1; |5 - 2| = 3 > 1) \quad |N| = 2 < 3$$

$$N(6) = \{5, 6, 7\} (1, 1 \leq 1) \quad |N| = 3 \geq 3 \text{ core}$$

$$N(7) = \{6, 7\} (|7 - 6| = 1; |7 - 15| = 8 > 1) \quad |N| = 2 < 3$$

$$N(15) = \{15\} (|15 - 7| = 8 > 1) \quad |N| = 1 < 3$$

DBSCAN — border and noise

0: $d(0, 1) = 1 \leq 1$, 1 core $\Rightarrow$ border. 2: $d(2, 1) = 1 \Rightarrow$ border. 5: $d(5, 6) = 1 \Rightarrow$ border. 7: $d(7, 6) = 1 \Rightarrow$ border. 15: nearest core 6 at $9 > 1 \Rightarrow$ noise.

DBSCAN — clusters

Cores 1 and 6: $d(1, 6) = 5 > 1$, separate chains. Clusters $\{0, 1, 2\}$ and $\{5, 6, 7\}$; noise $\{15\}$.

**Worked example 6.2 — cost**

$n = 1,000$.

$$\binom{1000}{2} = \frac{1000 \times 999}{2} = \frac{999,000}{2} = \mathbf{499,500} \text{ distances}$$

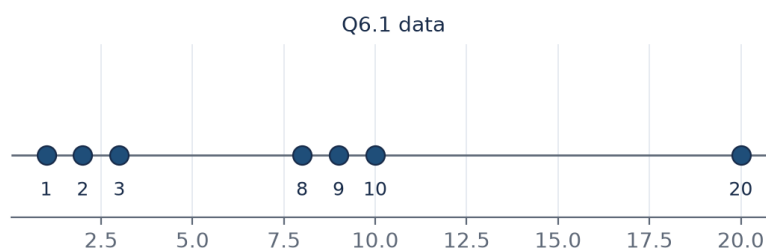
k-Means, $k = 5$, 20 iterations:

$$n \times k \times \text{iterations} = 1000 \times 5 \times 20 = 5000 \times 20 = \mathbf{100,000} \text{ distances}$$

Ratio $499,500/100,000 = 4.995 \approx 5$.

Questions

Q6.1 Points 1, 2, 3, 8, 9, 10, 20. **(a)** k-Means, $k = 2$, initial centroids 1 and 20: final centroids, partition, inertia. **(b)** Single linkage: merge table, $k = 2$ and $k = 3$ cuts. **(c)** DBSCAN, $\varepsilon = 1.5$, MINPTS = 3: core, border, noise, clusters. **(d)** Which method(s) give $\{1, 2, 3\}$ and $\{8, 9, 10\}$ as separate groups with 20 in neither?



Solution.

(a)

Assign: $1, 2, 3, 8, 9, 10 \rightarrow c_1$; $20 \rightarrow c_2$. Update $c_1 = \frac{33}{6} = 5.5$, $c_2 = 20$. Reassign: unchanged. Inertia = $\sum (x - 5.5)^2 = 20.25 + 12.25 + 6.25 + 6.25 + 12.25 + 20.25 = \mathbf{77.5}$. Partition $\{1, 2, 3, 8, 9, 10\}, \{20\}$.

(b)

$(1, 2)@1$, $(3, \{1, 2\})@1$, $(8, 9)@1$, $(10, \{8, 9\})@1$, $(\{1, 2, 3\}, \{8, 9, 10\})@5$, $(20, \text{rest})@10$.
 $k = 2$: $\{1, \dots, 10\}, \{20\}$. $k = 3$: $\{1, 2, 3\}, \{8, 9, 10\}, \{20\}$.

(c)

$N(1) = \{1, 2\}$ (2), $N(2) = \{1, 2, 3\}$ (3), $N(3) = \{2, 3\}$ (2), $N(8) = \{8, 9\}$ (2), $N(9) = \{8, 9, 10\}$ (3), $N(10) = \{9, 10\}$ (2), $N(20) = \{20\}$ (1). Core $\{2, 9\}$; border $\{1, 3, 8, 10\}$; noise $\{20\}$. Clusters $\{1, 2, 3\}, \{8, 9, 10\}$.

(d)

DBSCAN only.

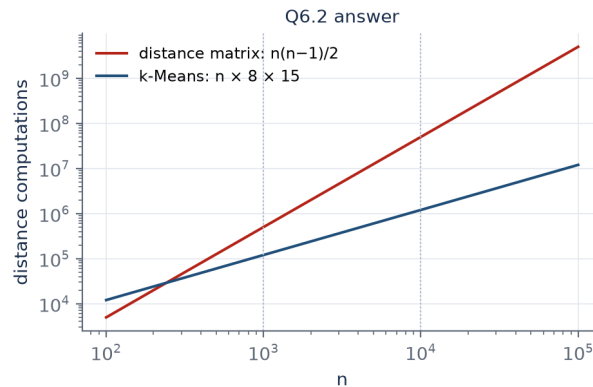


Q6.2 For $n = 10,000$: pairwise distances in a full matrix, and k-Means distances for $k = 8$ over 15 iterations. Growth factor of each relative to $n = 1,000$ (499,500 and 100,000).

Solution.

$$\binom{10000}{2} = \frac{10000 \times 9999}{2} = \mathbf{49,995,000}, \quad 10000 \times 8 \times 15 = \mathbf{1,200,000}$$

$$\frac{49,995,000}{499,500} = \mathbf{100.09}, \quad \frac{1,200,000}{100,000} = \mathbf{12}$$



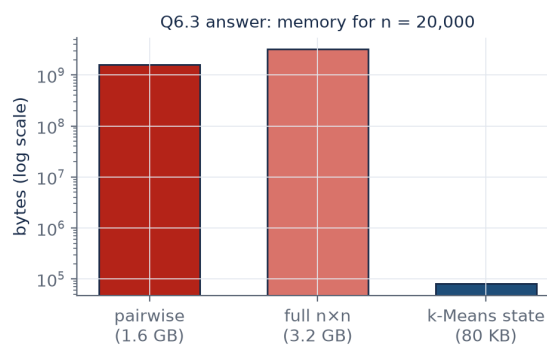
Q6.3 $n = 20,000$. (a) Number of pairwise distances and their size in bytes (8 bytes each) and GB (10^9 bytes). (b) Size of the full $n \times n$ matrix in GB. (c) k-Means with $k = 6$, $d = 10$: bytes for $k \times d$ centroids (8 bytes each) plus one 4-byte label per point; ratio of (a) to (c).

Solution.

(a)
 $\binom{20000}{2} = \mathbf{199,990,000}$; $\times 8 = 1,599,920,000$ bytes = **1.60 GB**.

(b)
 $20000^2 \times 8 = 3,200,000,000$ bytes = **3.2 GB**.

(c)
 $6 \times 10 \times 8 = 480$; $20000 \times 4 = 80,000$; total **80,480** bytes. Ratio $1,599,920,000 / 80,480 = \mathbf{19,880}$.



Q6.4 Three runs on one 500-point dataset:

Result 1: 4 labels; all 500 points labelled.

Result 2: 6 labels, plus 23 points labelled -1 .

Result 3: a table of 499 rows, each with two cluster ids and a height.

Match each to k-Means, hierarchical or DBSCAN. For Result 2 give the noise fraction; for Result 3 state what the row count equals.

Solution. Result 1: k-Means. Result 2: DBSCAN; noise fraction $23/500 = 4.6\%$. Result 3: hierarchical; $499 = n - 1$.